

ONTHATILE MONARENG

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## PROFILE

Computer Science student with hands-on experience building cybersecurity tools, backend applications, and real-time monitoring systems. Proficient in Python, Java, SQL, Linux, Git, FastAPI, and cloud technologies, with practical experience in vulnerability assessment, intrusion detection, log analysis, REST APIs, authentication systems, and backend development. AWS Certified Cloud Practitioner and Google Cybersecurity Professional Certificate holder. Seeking opportunities in software development, cybersecurity, IT, and related technical roles.

## PROJECTS

### Hercules Security SaaS – Cybersecurity platform

Python, FastAPI, SQL, Rest APIs, Linux

- Architected a scalable cybersecurity SaaS platform for automated vulnerability assessment.
- Implemented user authentication, email verification, password reset, MFA, API keys, team functionality, target verification, scan management and scan history.
- Engineered modular scanning engines to identify and classify vulnerabilities, utilising asynchronous processing to handle concurrent multi-tenant scan isolation.
- Integrated scheduled scanning and billing infrastructure as part of the platform architecture.

### Real-Time Intrusion Detection & Prevention System (Python)

Python, Linux, Security Monitoring

- Developed a real-time log monitoring engine using sliding window anomaly detection to identify suspicious traffic patterns.
- Built behavioural risk scoring algorithms to detect burst traffic and directory scanning, and other suspicious activity.
- Implemented automated IP blocking simulation and rate limiting simulations to validate system resilience against brute force attacks.
- Developed a live SOC-style dashboard using Rich, for real time visibility into detected threats and system activity.

- Implemented structured alert logging for auditing and incident tracking, supporting incident tracking and audit reporting.

### Web Log Intrusion Detection & Vulnerability Scanner

Python, Security Analysis, JSON

- Developed a rule-based intrusion detection system for identifying and classifying suspicious web traffic by severity.
- Implemented structured JSON reporting to facilitate seamless integration with external SIEM tools for centralised log analysis.
- Built scanning logic to detect suspicious web traffic patterns, including SQL injection and XSS attempts.
- Designed a system to analyse the web activity and provide actionable security information

### School Management System (Team Project)

Java, OOP, Backend Development

- Collaborated in backend development of administrative web system, implementing role based access control to manage student data integrity.
- Implemented CRUD operations for student registration, optimising database queries to reduce record retrieval latency.
- Contributed to system organization, application logic and system structure

## **TECHNICAL SKILLS**

Languages: Python, Java, SQL, HTML/CSS

Cybersecurity: Intrusion detection, Vulnerability Scanning, Log Analysis, Risk Assessment, Security Monitoring

Backend & APIs: FastAPI, REST APIs, JSON, OOP

Cloud: AWS IAM, S3, Google Cloud

Tools & Systems: Linux, Windows, Git, Android Studio

Core Strengths: Debugging, log analysis, problem solving, teamwork, system analysis

## **EDUCATION**

BSc Computer Science — Regenesys University (2025 – 2027)

NSC (Matric) — Maragon Mooikloof (2024)

## **CERTIFICATIONS**

- AWS Certified Cloud Practitioner
- Google Cybersecurity Professional Certificate